

project to project and according to the applicability of GIS in different industries. Broadly speaking, there are three distinct work areas: data conversion, GIS Application Development and GIS Analysis. Let's look at each stage in relation with the work areas to understand the processes and people responsible for getting a GIS project up and running.

Where's my data?

Data input, both spatial and statistical, for a GIS project varies according to industry requirements and the company for which the GIS solution is being customised. Data can be sourced from existing digital maps, satellite remote

needed. A background in mathematics would help you with the mathematical analysis in GIS projects."

Programmed to customise

GIS application developers or programmers of varying skills are required on spatial projects for coding applications or customisation after a map is digitised by a CAD operator for graphical display. Customisation can be in the form of creating menus or algorithms to find the shortest route between two points to facilitate easy decision making for users.

To become a developer, you need to have a background in computers with thorough

knowledge of software engineering tools and programming languages such as Visual Basic, C, C++, Visual C++, COM, EJB and JSP. For complex applications, you must have a knowledge of MDL (for Microstation),

AutoLISP (for Autodesk), Arc Macro Language (AML) in ARC/INFO (for ESRI platforms), Avenue in ArcView XML and GML or software such as MapObjects or ArcSDE. Knowledge of GIS fundamentals and familiarity with spatial databases and database systems is essential as is awareness of algorithms and data structures.

"GIS application development is getting diversified into two areas: Application GIS and IT-enabled GIS," says Yogendran S., a GIS analyst at IIC Technologies, Hyderabad. "IT professionals can easily make the transition to IT enabled GIS irrespective of their field. A fair knowledge of GIS fundamentals and software such as ArcInfo, MapInfo and GeoMedia will serve as an easy passport for software professionals into GIS."

Designs on data

A GIS closely integrates two databases: a spatial database containing coordinates related to location and an attribute (statistical) database comprising, for instance, demographic data or statistics about a particular location. Database design includes the planning and development of a database for storage of spatial and non-spatial data attributes, data structures, linking the database model to the application model, and so on. Database building involves different spatial database structures (both vector and raster), topology, search algorithms and trees (binary and spatial). The role of the data

Location-based Services

You're driving through an unfamiliar area when your car suddenly breaks down. You get your mobile phone out and, through the use of interactive maps, access a database which tells you where the nearest service station is. This is one of the ways in which LBS or Location Based Services work.

The elements involved in LBS technology include positioning, GIS and location management function. The 'positioning' element is similar to GPS, except that network positioning is used to determine the location of a mobile device.

GIS systems are used to map the geographical data around the position of the user, and to map the locations of the places and services that a user might need.

The location management function is used to process the location of a mobile device and the GIS data around it. It acts as a gateway between the positioning equipment and the LBS infrastructure.

The LBS application on a user's handset could work with other location technologies to determine the user's location and provide a list of services within a certain proximity to the user.

LBS technology is used mostly for personalised customer service, which could extend into location based information and billing—a user can be charged different airtime rates relative to whether he is at home or office. LBS can also be used in fleet management systems to track the position of vehicles. Emergency services is another area where there's tremendous demand for LBS.



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GIS analyst, IIC Technologies, Hyderabad

sensing, photogrammetry, purchase of databases from commercial data vendors, surveys or even paper maps. In India, physical data isn't readily available or accessible. Besides, most of the data is not available in a digitised form. Hence, a chunk of the manpower involved in a GIS project is involved in digitising data and data conversion—processing and digitising images from remote sensing and aerial photos, map digitisation or digitising manual records. Map digitisation involves scanning or digitising existing paper maps to a vector or a raster format, which can then be loaded onto a GIS platform.

Digitisation is a low-end job for which you need proficiency in CAD software such as AutoCad, AutoLisp or Microstation. "Map digitisation is more labour-intensive and the technical skills and domain knowledge required to do the job are comparatively limited," says Maneesh Prasad, Director of Gisdevelopment.net. "The pay is not high and one can become a supervisor, a data editor, or be in charge of QC (Quality Control). If you're involved in digitisation, you should enhance your skills as it will give you more options for career advancement," adds Prasad. "Diploma engineers fresh from college with skills in Microstation, or GIS software such as Arcinfo or Geomedia can opt for data conversion," says Manoj Thakur, president and CEO of Nucleus Netsoft & GIS India. "As far as aptitude is concerned, an analytical mind and the ability to extract information from maps are